04-DEVELOPMENT-ROADMAP.md

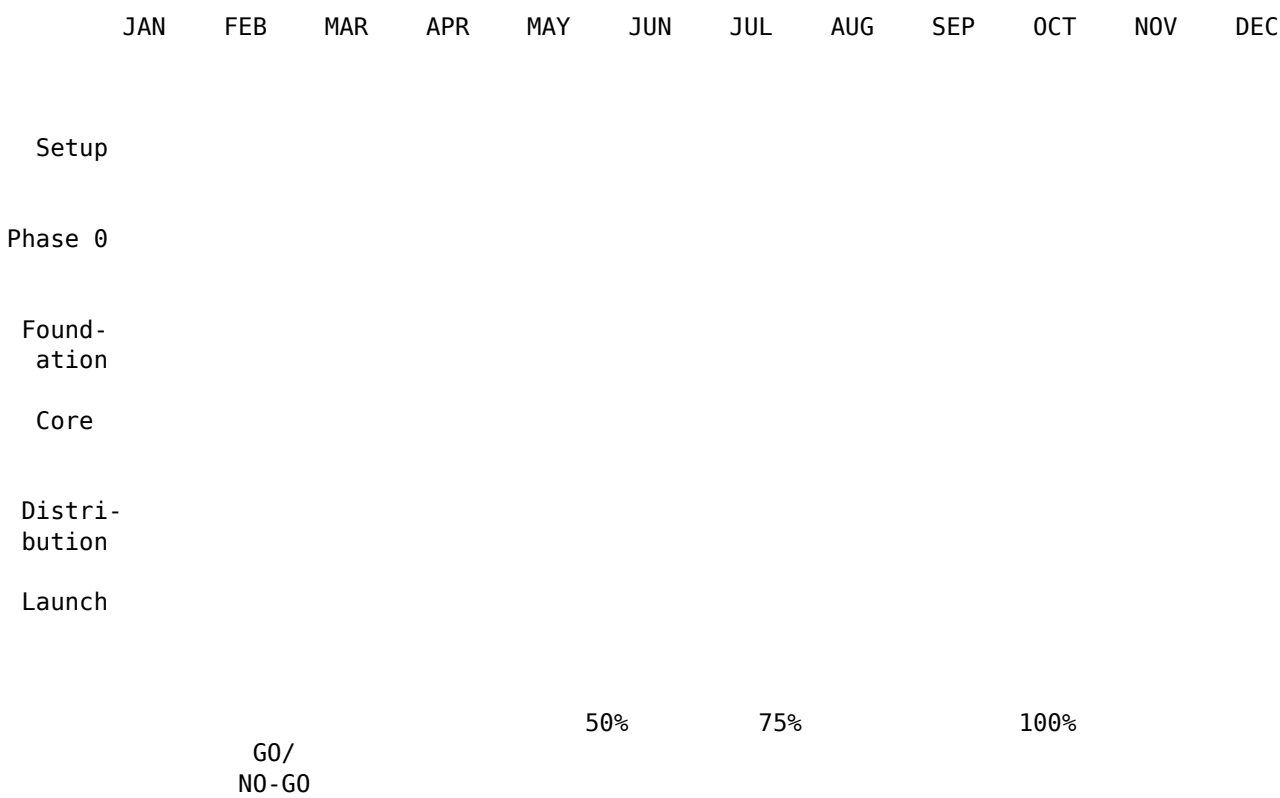
.things that Matter

Development Roadmap & Milestones

Document Information	
Company Name	.things that Matter
Document Type	Development Roadmap
Version	1.0
Date	January 21, 2026
Author	Matome Mboweni, CTO

1. Roadmap Overview

1.1 Phase 1 Timeline (2026)



1.2 Sprint Structure

Sprint	Duration	Focus
Setup	2 weeks	Environment, contracts, planning
Phase 0	4 weeks	Steganography validation
Foundation	8 weeks	FlatBuffers, Rust compiler
Core	12 weeks	CLIP, signing, steganography engine
Distribution	8 weeks	CDN, API, browser extension
Launch	8 weeks	Beta merchants, public release

2. Detailed Sprint Breakdown

2.1 Sprint 0: Setup (Jan 21 - Feb 2, 2026)

\1 2 weeks
\1 Establish foundation for development

Week	Tasks	Deliverables
Week 1	Sign new company contract	Signed agreement
	Set up development environment	Cursor Pro, VS Code, Rust toolchain
	Create GitHub organization	thingsthatmatter org
	Initialize monorepo structure	packages/ directory
Week 2	Set up Supabase project	Database schema
	Configure Vercel deployment	CI/CD pipeline
	Set up Cloudflare R2 bucket	Storage ready
	Begin Rust learning	Chapters 1-5 of The Rust Book

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- [ ] Contract signed
- [ ] Development environment operational
- [ ] Repository initialized with structure
- [ ] Infrastructure accounts created

2.2 Sprint 1: Phase 0 - Validation (Feb 3 - Mar 2, 2026)

\1 4 weeks
\1 Prove steganography viability
\1 R2,000

Week	Tasks	Deliverables
Week 3	Research steganography techniques	Technical notes
	Build basic XMP injection tool (Rust)	xmp_inject.rs
	Build EOF append tool (Rust)	eof_append.rs
	Test on 5 platforms	Initial results
Week 4	Test on 15 additional platforms	Full platform matrix
	Document survival rates	Platform compatibility table
	Build basic extraction tool	extract_payload.py
	CLIP API integration test	clip_test.py
Week 5	Build end-to-end prototype	JSON → .thing → image → decode
	Cost analysis for CLIP	Cost projections
	Validate Ed25519 signing	sign_verify.py
Week 6	Compile results	Phase 0 Report
	Go/No-Go decision	Decision document
	Plan adjustments if needed	Updated roadmap

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Platform	Test	Status
Discord	Upload/download image	Pending
Telegram	Send/receive image	Pending
Twitter/X	Tweet with image	Pending
Instagram	Post image	Pending
Facebook	Share image	Pending
WhatsApp	Send image	Pending
Signal	Send image	Pending

Reddit	Post image	Pending
Pinterest	Pin image	Pending
LinkedIn	Post image	Pending
Slack	Share image	Pending
Email (Gmail)	Attach image	Pending
Email (Outlook)	Attach image	Pending
Google Photos	Upload	Pending
iCloud Photos	Upload	Pending
Dropbox	Upload	Pending
Google Drive	Upload	Pending
OneDrive	Upload	Pending
Direct download	HTTP transfer	Pending
S3/R2	CDN delivery	Pending

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Survival Rate	Decision
≥70%	Full steam ahead with steganography
50-69%	Hybrid approach: steg + matter:// URIs
<50%	Pivot to matter:// as primary distribution

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- [ ] 20 platforms tested
- [ ] Survival rate documented
- [ ] End-to-end prototype working
- [ ] Go/No-Go decision made
- [ ] Phase 0 Report complete

2.3 Sprint 2-3: Foundation (Mar 3 - Apr 27, 2026)

\1 8 weeks
\1 Build compiler foundation

Sprint 2: FlatBuffers & Schema (Mar 3 - Mar 30)

Week	Tasks	Deliverables
Week 7	Design FlatBuffers schema	thing.fbs
	Set up flatc compiler	Build scripts
	Generate Rust bindings	thing_generated.rs
	Write schema documentation	Schema docs
Week 8	Implement basic serializer (Rust)	serialize.rs
	Implement basic deserializer (Rust)	deserialize.rs
	Unit tests for serialization	tests/serialize.rs
	Zstd compression integration	compress.rs

Sprint 3: Rust Compiler Foundation (Mar 31 - Apr 27)

Week	Tasks	Deliverables
Week 9	Product JSON validation (serde_json + validator)	validator.rs
	Image download utility	fetch_image.rs
	CLIP API integration	clip_client.rs
	Vector embedding pipeline	vectorize.rs
Week 10	Ed25519 signing implementation	crypto.rs
	Signature verification	verify.rs
	Key management utilities	keys.rs

	Integration tests	tests/crypto.rs
Week 11	Full compile() function	compiler.rs
	CLI tool	matter-cli (Rust binary)
	Batch processing support	batch.rs
	Error handling & logging	errors.rs
Week 12	End-to-end testing	tests/e2e.rs
	Performance benchmarking	benches/
	Documentation	compiler_docs.md
	Code review & cleanup	Refactored code

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- [ ] FlatBuffers schema finalized
- [ ] Rust compiler produces valid .thing files
- [ ] CLIP vectors embedded correctly
- [ ] Ed25519 signatures verify correctly
- [ ] CLI tool operational
- [ ] <500ms compile time achieved

2.4 Sprint 4-6: Core (Apr 28 - Jul 20, 2026)

\1 12 weeks
\1 Complete core functionality

Sprint 4: Steganography Engine (Apr 28 - May 25)

Week	Tasks	Deliverables
Week 13	XMP metadata injection library	xmp.rs
	EOF binary append library	eof.rs
	Image format handling (JPEG/PNG)	image_utils.rs
	Polyglot image generator	polyglot.rs
Week 14	XMP beacon format specification	xmp_spec.md
	Gist JSON structure	gist.rs
	Metadata extraction library	extract.rs
	Unit tests	tests/steg.rs
Week 15	Integration with compiler	Updated compiler.rs
	Output: .thing + polyglot image	Full pipeline
	Quality validation	image_quality.rs
	Documentation	steg_docs.md
Week 16	Platform-specific optimizations	platform_compat.rs
	Fallback strategies	fallback.rs
	Error recovery	recovery.rs
	Performance tuning	Optimized code

Sprint 5: Shadow Index API (May 26 - Jun 22)

Week	Tasks	Deliverables
Week 17	API design & OpenAPI spec	openapi.yaml
	Supabase schema (hash → metadata)	migrations/
	POST /compile endpoint	api/compile.ts
	GET /resolve endpoint	api/resolve.ts
Week 18	GET /gist endpoint	api/gist.ts
	POST /batch endpoint	api/batch.ts
	Authentication (API keys)	auth.ts
	Rate limiting	ratelimit.ts

Week 19	Cloudflare R2 integration	storage.ts
	Upload pipeline	upload.ts
	CDN URL generation	cdn.ts
	Caching layer (Redis)	cache.ts
Week 20	Error handling	errors.ts
	Logging & monitoring	logging.ts
	API documentation	api_docs.md
	Deploy to Vercel	Production API

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Sprint 6: matter:// Protocol (Jun 23 - Jul 20)

Week	Tasks	Deliverables
Week 21	URI scheme specification	matter_uri_spec.md
	Hash encoding (Base58)	encoding.ts
	URI generation in compiler	Updated compiler
	URI parsing library	uri_parser.ts
Week 22	Protocol handler research	Browser compatibility notes
	Fallback HTTPS URLs	fallback_urls.ts
	Deep linking support	deeplink.ts
	Documentation	protocol_docs.md
Week 23	Rust: Steganography integration	matter-core/src/steg/
	Rust: CLIP client optimization	matter-core/src/clip/
	WASM build optimization	Smaller WASM
	Performance benchmarks	benches/
Week 24	Rust: Full pipeline optimization	matter-core/src/compiler.rs
	Integration tests	tests/integration.rs
	Documentation	rust_docs.md
	Code review & refactoring	Optimized codebase

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- [ ] Steganography engine complete
- [ ] Shadow Index API deployed
- [ ] matter:// URI generation working
- [ ] Rust port 30% complete
- [ ] All core features functional

2.5 Sprint 7-8: Distribution (Jul 21 - Sep 14, 2026)

\1 8 weeks
\1 Browser extension and distribution infrastructure

Sprint 7: Browser Extension (Jul 21 - Aug 17)

Week	Tasks	Deliverables
Week 25	Extension architecture design	Architecture doc
	Manifest v3 setup	manifest.json
	Content script: DOM observer	content.js
	XMP beacon detection	detect.js
Week 26	Background worker: fetch .thing	background.js
	WASM: libmatter compilation	libmatter.wasm
	Signature verification in WASM	verify.wasm
	IndexedDB caching	cache.js
Week 27	Overlay UI: Shadow DOM	overlay.js
	Product card component	card.js

	Styling (isolated CSS)	styles.css
	"Buy Now" flow	checkout.js
Week 28	matter:// URI interception	protocol.js
	Settings page	settings.html
	Error handling	errors.js
	Testing & debugging	test_extension.js

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Sprint 8: Polish & Integration (Aug 18 - Sep 14)

Week	Tasks	Deliverables
Week 29	Extension performance optimization	Optimized code
	Bundle size reduction	<100KB target
	Cross-browser testing (Chrome, Edge)	Compatibility report
	Accessibility review	A11y fixes
Week 30	Integration: compiler + API + extension	Full pipeline test
	End-to-end testing	E2E test suite
	Bug fixes	Issue resolution
	Documentation	User guide
Week 31	Rust port: steganography	steg.rs
	Rust port: CLIP client	clip.rs
	WASM build optimization	Smaller WASM
	Performance benchmarks	Benchmark report
Week 32	Security review (self)	Security checklist
	Code cleanup	Refactored code
	README updates	Documentation
	Prepare for beta	Beta checklist

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- [ ] Browser extension functional
- [ ] End-to-end demo working
- [ ] Rust compiler 50-70% complete
- [ ] WASM verification working
- [ ] Ready for beta testing

2.6 Sprint 9-10: Launch (Sep 15 - Dec 15, 2026)

\1 13 weeks
\1 Beta launch and public release

Sprint 9: Beta Program (Sep 15 - Oct 26)

Week	Tasks	Deliverables
Week 33	Beta merchant onboarding (5-10)	Merchant accounts
	Merchant documentation	Onboarding guide
	Support process setup	Support workflow
	Feedback collection system	Feedback forms
Week 34	Compile merchant catalogs	1000+ products
	Monitor performance	Metrics dashboard
	Bug triage & fixes	Bug fixes
	Merchant feedback calls	Feedback notes
Week 35	Iterate on feedback	Updated features
	Extension improvements	v0.2 release
	API improvements	API updates
	Documentation updates	Updated docs

Week 36-37	Continue beta testing	Ongoing support
	Stability improvements	Stable release
	Performance tuning	Optimized system
	Prepare Chrome Web Store	Store assets

Sprint 10: Public Launch (Oct 27 - Dec 15)

Week	Tasks	Deliverables
Week 38	Chrome Web Store submission	Submitted extension
	Review response handling	Updates if needed
	Landing page	thingsthatmatter.com
	Documentation site	docs.thingsthatmatter.com
Week 39-40	Marketing preparation	Blog posts, demos
	Developer documentation	SDK docs
	API documentation	API reference
	Support documentation	FAQ, troubleshooting
Week 41-42	Public launch	Live!
	Monitor adoption	Analytics
	Support incoming users	User support
	Bug fixes	Patch releases
Week 43-44	Post-launch stabilization	Stable v1.0
	Retrospective	Lessons learned
	Phase 2 planning	Phase 2 roadmap
	Year-end wrap	Annual report

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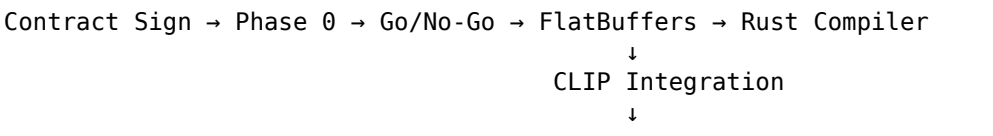
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- [ ] 5-10 beta merchants onboarded
- [ ] 1000+ products compiled
- [ ] Extension in Chrome Web Store
- [ ] Documentation complete
- [ ] Public launch successful

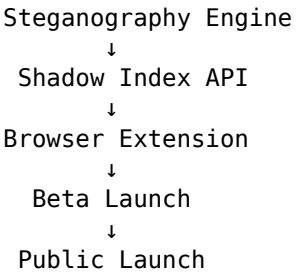
3. Milestone Summary

Milestone	Date	Completion	Key Deliverable
Contract Signed	Jan 21, 2026	0%	New company agreement
Phase 0 Complete	Mar 2, 2026	15%	Steganography validation
Foundation Complete	Apr 27, 2026	30%	Rust compiler MVP
Core Complete	Jul 20, 2026	50%	Full pipeline working
75% Milestone	Aug 17, 2026	75%	Browser extension functional
Beta Launch	Oct 26, 2026	90%	Merchants onboarded
Public Launch	Dec 15, 2026	100%	Chrome Web Store live

4. Dependencies & Critical Path

4.1 Critical Path





4.2 Dependency Matrix

Task	Depends On	Blocks
Phase 0	Contract signed	Go/No-Go decision
FlatBuffers schema	Phase 0 Go	Rust compiler
Rust compiler	FlatBuffers	Steganography
CLIP integration	API account	Rust compiler
Ed25519 signing	None	Compiler
Steganography	Rust compiler	Shadow Index
Shadow Index API	Steganography	Browser extension
Browser extension	Shadow Index, WASM	Beta launch
Rust optimization	Rust compiler working	Performance tuning
Beta launch	Extension, merchants	Public launch

5. Risk to Timeline

Risk	Probability	Impact	Mitigation	Schedule Impact
Phase 0 fails (<50% survival)	Medium	High	Pivot to matter://	+2 weeks
Cursor Pro limits slow dev	Medium	Medium	Batch work	+2-4 weeks
CLIP API pricing changes	Low	Medium	Self-host GPU	+4 weeks
Chrome rejects extension	Low	High	Firefox first	+4 weeks
Rust learning takes longer	Low	Low	C/C++ background accelerates learning	None
Merchant adoption slow	Medium	Medium	Developer focus	Shift strategy

6. Resource Allocation

6.1 Time Allocation by Category

Category	Hours/Week	% of Time
Core development	20	67%
Learning (Rust)	3	10%
Documentation	2	7%
Testing	3	10%
Admin/meetings	2	6%
Total	**30**	**100%**

6.2 Monthly Budget Burn

Month	Amount	Cumulative	Key Expenses
Feb	R8,000	R8,000	Phase 0, setup
Mar	R6,500	R14,500	Development tools
Apr	R6,000	R20,500	CLIP API

May	R6,000	R26,500	CLIP, storage
Jun	R6,300	R32,800	API hosting
Jul	R6,000	R38,800	Development
Aug	R6,200	R45,000	Extension dev
Sep	R6,000	R51,000	Testing
Oct	R6,000	R57,000	Beta support
Nov	R7,500	R64,500	Launch prep
Dec	R7,500	R72,000	Public launch

7. Progress Tracking

7.1 Weekly Checkpoints

Every Friday:
- [ ] Code committed to repository
- [ ] Progress update to CEO
- [ ] Blockers identified
- [ ] Next week planned

7.2 Monthly Demos

First Monday of each month:
- [ ] Working demo of new features
- [ ] Metrics review
- [ ] Budget review
- [ ] Timeline adjustment if needed

7.3 Milestone Reviews

At each milestone:
- [ ] Deliverables verified
- [ ] Quality assessment
- [ ] Go/No-Go for next phase
- [ ] Retrospective

8. Success Criteria by Phase

Phase 0 Success (Feb 2026)

- [ ] ≥50% steganography survival rate
- [ ] CLIP integration working
- [ ] End-to-end prototype functional
- [ ] Clear Go/No-Go decision

Foundation Success (Apr 2026)

- [ ] FlatBuffers schema finalized
- [ ] Rust compiler producing valid .thing files
- [ ] <500ms compile time
- [ ] CLI tool operational

Core Success (Jul 2026)

- [ ] Steganography engine complete
- [ ] Shadow Index API deployed
- [ ] matter:// URIs generating
- [ ] Rust port 30% complete

Distribution Success (Sep 2026)

- [ ] Browser extension functional
- [ ] End-to-end demo working
- [ ] WASM verification working
- [ ] Ready for beta

Launch Success (Dec 2026)

- [ ] 5-10 merchants onboarded
- [ ] 1000+ products compiled
- [ ] Extension in Chrome Web Store
- [ ] Public launch complete

\1 1.0
\1 January 21, 2026
\1 Monthly